

Temporal Topic Modeling and Citation Recommendation for Scientific Literature

CS6485: Capstone Project II

Presented By: Khushbu Chaudhary

Roll number: CS24MTECH14012

Guided by: Dr. Srijith P.K.



భారతీయ సాంకేతిక విజ్ఞాన సంస్థ హైదరాబాద్
भारतीय प्रौद्योगिकी संस्थान हैदराबाद
Indian Institute of Technology Hyderabad

- Motivation
- Project Objectives
- Literature Survey
- Foundational Hawkes Theory
- Proposed Framework & Architecture
- Dataset & Preprocessing Pipeline
- Experimental Setup & Testing Harness
- Results
- Future Work

Motivation

- Scientific publications in AI and ML are growing at an unprecedented rate.
- Researchers face two key challenges:
 - Tracking emerging and declining research topics
 - Identifying the most relevant prior work for citation
- Traditional topic models such as LDA provide only static topic distributions.
- They fail to capture how influential papers trigger future research activity over time.
- Scientific publishing exhibits a self-exciting behavior, where one impactful paper leads to cascades of related work.
- Hawkes processes are well-suited for modeling these temporal influence patterns.

Objective

Develop a citation-aware temporal modeling framework capable of:

- Predicting future research trends
- Modeling topic evolution over continuous time
- Capturing influence propagation between scientific papers
- Identifying semantically relevant prior work using attention mechanisms

Bridge topic modeling and temporal influence learning for scientific trend prediction.

Literature Survey - Hawkes Topic Model (HTM)

- HTM combines topic modeling with multivariate Hawkes processes.
- It jointly learns:
 - latent topics
 - influence relationships between authors or nodes
- Uses exponential temporal decay to model event triggering.
- **Strengths:** Captures diffusion and influence dynamics, Strong probabilistic foundation
- **Limitations (for our task):**
 - Focused on influence discovery, not future topic prediction
 - Exponential decay cannot model delayed citation peaks
 - Limited semantic understanding due to lack of attention mechanisms

Literature Survey - Dynamic Structured Neural Topic Model

- Neural topic model that captures topic evolution across time.
- Uses self-attention to model relationships between topics.
- Incorporates citation information through citation regularization.
- **Strengths:** Handles dynamic topic evolution, Learns time-varying topic-word distributions, Achieves strong topic coherence and perplexity performance
- **Limitations:**
 - Uses discrete time bins, losing fine-grained temporal dynamics
 - Does not explicitly model cascading influence behavior
 - Joint optimization increases training complexity

Introduction to Hawkes Processes

- A Hawkes process is a self-exciting point process. The key idea: past events increase the probability of future events.
- Intuitive Example: A Twitter Cascade
 - **The Base Rate μ :** The background noise. Even on a quiet day, users post tweets at a low, steady baseline frequency.
 - **The Triggering Event:** A high-profile user posts a breaking piece of news. This acts as a shock to the system.
 - **Self-Excitation:** Cascading waves; each reply acts as a new event sparking subsequent retweets.
 - **Temporal Decay:** As time passes, user interest fades, and the probability of new posts decays back down toward the background base rate μ .

Mathematical Formulation of the Hawkes Process

- **The Conditional Intensity Function $\lambda_k(t)$:**

$$\lambda_k(t) = \mu_k + \sum_{e: t_e < t} \alpha_{e,k} \cdot f(t - t_e)$$

- Variable Breakdown:
- $\lambda_k(t)$: The active continuous intensity (popularity) of topic k at fractional timestamp t .
- μ_k : The stable, background base rate of occurrence for topic k .
- $\sum_{e: t_e < t}$: The historical summation gathering the ripples of influence from all past events e that arrived prior to the current evaluation point.
- $\alpha_{e,k}$: The excitation weight parameter, determining how strongly past document e activated topic k .
- $f(t - t_e)$: The temporal decay kernel, downweighting the influence of past events as the chronological time gap increases.

Limitations of Existing Approaches

- **Standard Hawkes Process:**

- Uses exponential decay, assuming influence is strongest immediately after publication.
- Cannot capture delayed citation behavior commonly observed in research papers.
- Treats past events uniformly, without understanding semantic relevance between papers.
- Does not explicitly model topic-level representations or evolving research themes.

- **DSNTM:**

- Divides time into discrete intervals, losing fine-grained temporal information.
- Focuses mainly on topic evolution rather than influence propagation.
- Lacks an explicit self-exciting mechanism to model citation cascades.
- Joint learning of topics and temporal dynamics increases model complexity.

Architecture Design Rationale

- **Weibull Temporal Kernel:** Models delayed citation peaks more effectively than standard exponential decay.
- **Topic-Space Attention Mechanism:** Learns which past papers are semantically relevant, not just temporally recent.
- **Frozen LDA Topic Representation:** Separates topic discovery from temporal influence modeling, improving interpretability and training stability.
- **Citation-Supervised Attention:** Uses citation information to guide attention weights toward real influence patterns.
- **Continuous-Time Modeling:** Preserves exact publication timing without relying on discrete time bins.

Architecture Flowchart

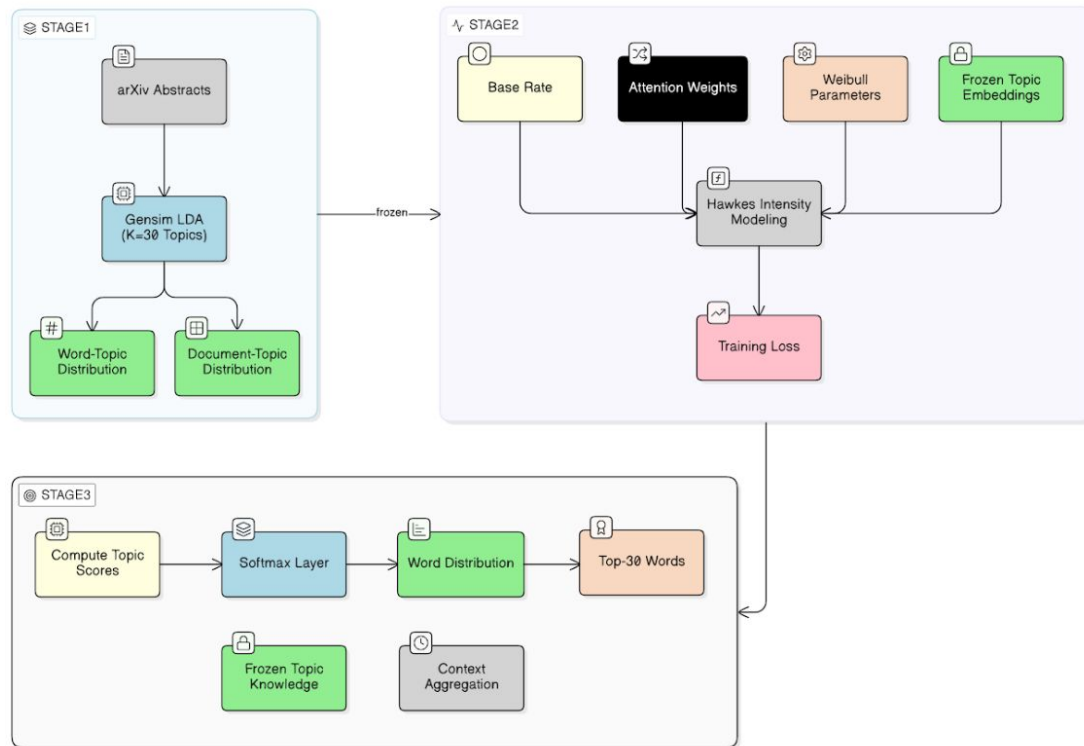


Fig. 1: Architecture

Proposed Methodology

- **Frozen LDA Topic Representation:** LDA is used only for discovering topic distributions, while Hawkes modeling handles temporal evolution separately.
- **Attention-Based Influence Modeling:** The attention mechanism learns which past papers are most relevant to the current research context.
- **Topic-Specific Influence Learning:** Different research areas influence future work differently, enabling asymmetric topic interactions.
- **Context Vector $c(T)$ for Prediction:** Future papers are represented using the recent topic context of the field rather than unavailable future document embeddings.
- **Consistent Training and Inference:** Using contextual topic representations ensures stable behavior during both training and future prediction.

Intensity function

- The intensity of topic k at prediction time T^* is:

$$\lambda_k(T^*) = \mu_k + \sum_{e: t_e < T^*} \underbrace{\theta_e[k]}_{\text{topic weight}} \cdot \underbrace{A_{e, \text{ctx}}}_{\text{attention}} \cdot \underbrace{f_{\Delta}(T^* - t_e)}_{\text{Weibull decay}}$$

- Each past paper e contributes through three multiplicative factors:
- $\theta_e[k]$: How much paper e is about topic k (from frozen LDA). An NLP paper contributes strongly to NLP topics, weakly to vision topics.
- $A_{e, \text{ctx}}$: How relevant paper e is to the current research context (learned via attention). Not all past papers matter equally.
- $f_{\Delta}(T^* - t_e)$: How recent paper e is, with influence that can peak after a delay (Weibull kernel).

Attention Mechanism for Citation-Aware Topic Influence

- The attention module determines which past papers are most influential for future topic prediction.

$$A_{e, \text{ctx}} = \text{softmax}\left(\frac{\mathbf{c}(T^*)^\top \mathbf{W}_Q \cdot \mathbf{W}_K \boldsymbol{\theta}_e}{\sqrt{K}}\right)$$

- $\mathbf{c}(T^*)$: Context vector representing the current research landscape
- $\mathbf{W}_Q, \mathbf{W}_K$: Learnable query and key projection matrices
- K : Number of latent topics
- The attention can learn asymmetric relationships: "RL papers tend to cite optimization papers" even without direct topic overlap between query and key.
- The citation loss (BCE between attention weights and actual citation indicators) grounds this in real citation behavior.

Weibull Temporal Decay

- Standard exponential decay assumes that influence is strongest immediately after publication and decreases continuously over time.
- In real scientific publishing, citation influence often: rises gradually, reaches a peak after weeks or months, and then slowly declines.
- The Weibull kernel naturally captures this delayed influence behavior.

$$f_{\Delta}(\Delta t; k_w, \lambda_w) = \frac{k_w}{\lambda_w} \left(\frac{\Delta t}{\lambda_w} \right)^{k_w-1} \exp \left(- \left(\frac{\Delta t}{\lambda_w} \right)^{k_w} \right)$$

- where k_w = shape parameter ($k_w > 1$ gives delayed peak), λ_w = scale parameter (controls duration of influence)
- Citation influence usually peaks after some delay, whereas exponential decay decreases immediately from $t=0$. The Weibull kernel ($k_w > 1$) effectively models this rise–peak–decay behavior.

Training Objective and Loss Functions

- The model is trained using a multi-task objective that combines temporal event prediction and citation-aware attention learning
- Hawkes Loss ($\mathcal{L}_{\text{hawkes}}$) : Maximizes the likelihood of observed publication events.

$$\mathcal{L}_{\text{Hawkes}} = \sum_d \left[\sum_k \log \lambda_k(t_d) - \int_0^T \lambda_k(t) dt \right]$$

- 1st term: reward high intensity at observed events, 2nd term (compensator): penalize intensity between events.
- Citation Loss ($\mathcal{L}_{\text{citation}}$) : Aligns attention weights with the actual citation graph.

$$\mathcal{L}_{\text{cite}} = \sum_{d, e} \text{BCE}(A_{e,d}, C_{e,d})$$

- $C_{e,d} = 1$ if paper e cites paper d
- The total loss combines both objectives using a balancing parameter gamma.

$$\mathcal{L} = \mathcal{L}_{\text{Hawkes}} + \gamma \cdot \mathcal{L}_{\text{cite}}$$

- Joint optimization improves both temporal prediction and citation-aware relevance modeling.

Dataset & Preprocessing

Component	Description
Dataset Source	SciEvo dataset from HuggingFace
Corpus	arXiv papers from cs.CL and cs.LG
Time Range	2014 - 2021
Topic Modeling	LDA with (K = 30) topics
Topic Representation	Frozen after LDA training
Training Set	Papers published up to 2020
Test Set	Papers published in 2021

Table No. 1: SciEvo Dataset Specifications

Preprocessing pipeline

- Fetch abstracts + metadata from SciEvo / Semantic Scholar
- Text cleaning: lowercasing, URL/email/number removal, strip non-alphabetic characters
- Stopword removal: English stopwords + domain-specific stoplist ("paper," "method," "results," "propose," etc.)
- Lemmatization of all tokens
- Bigram collocation detection (e.g., "reinforcement_learning," "language_model")
- Train LDA ($K=30$) $\rightarrow$ produce frozen β (topic-word matrix) and θ (doc-topic matrix)
- Citation graph extraction: sparse matrix of paper-to-paper citations from the corpus
- Construct temporal event sequences: each paper becomes an event (time, topic vector, citation list)

Preprocessing pipeline

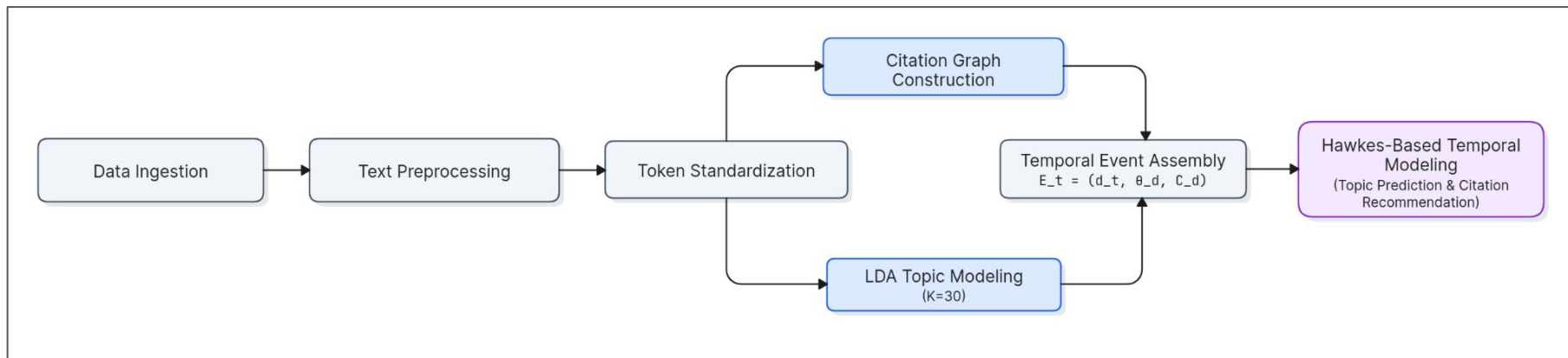


Fig. 2: Preprocessing Pipeline

Two Evaluation Modules

1. Topic Prediction:

- Predict next-year topic & word distributions
- Metrics: JSD, Cosine, Top-30 Overlap
- Perplexity, NPMI coherence
- Compared against: DSNTM, Static LDA

2. Citation Recommendation:

- Rank which past papers a new paper cites
- Metrics: AUC-ROC, MRR, Recall@K
- Continuous-time Hawkes intensity
- DSNTM cannot perform this task at all

Topic Prediction Results

Metric	Hawkes(Our)	DSNTM	Static LDA
Pred JSD	0.248	0.064	0.226
Pred Cosine	0.923	0.991	0.944
Top-30 Overlap	0.767	0.833	0.800
NPMI Coherence	0.055	0.019	-
Perplexity	2603	1613	-

Table No. 2: Topic Prediction Results

- DSNTM wins on word prediction (its time-varying β adapts vocabulary each year), but our model produces more coherent topics (NPMI 0.055 vs 0.019).

Citation Recommendation Results

Metric	Hawkes(Our)	Static LDA	Content+Recency	Uniform Hawkes
AUC-ROC	0.8151	0.7710	0.459	0.390
MRR	0.0448	0.0440	0.034	0.001
Recall@5	0.0055	0.0063	0.0048	0.00003
Recall@20	0.0160	0.0106	0.0115	0.0001

Table No. 3: Citation Recommendation Results

Citation Recommendation Results

- Our model achieves the best overall performance with the highest AUC-ROC (0.8151), showing that combining topic relevance with temporal dynamics and learned attention produces the strongest citation ranking.
- Uniform Hawkes collapses to near-random (0.390) recency alone cannot predict citations; content awareness is essential.
- Content+Recency (0.459) also underperforms-naive content matching without Hawkes-style influence propagation is insufficient.
- Static LDA is competitive on Recall@5 due to strong topic similarity, but our model overtakes at Recall@20 (0.0160 vs 0.0106) and also achieves better MRR (0.0448 vs 0.0440) by surfacing temporally influential papers that topic similarity alone misses.
- Low absolute Recall values across all models are expected given the large candidate pool (thousands of papers, each citing only ~20-40).

- Add cross-domain interaction weights to handle diffuse excitation across broad CS sub-fields (cs.AI, cs.LG, cs.CL).
- Replace linear intensity with a neural gating mechanism to better isolate sharp, sudden trend bursts and reduce forecasting JSD.
- Shift from frozen LDA to a Dynamic Topic Model (DTM) so word distributions evolve with research vocabulary over time.
- Learn paper-specific Weibull parameters (k_w, λ_w) instead of global ones, capturing unique citation lifespans per paper type.

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Thank You!